01 SYSTEM LAYOUT: FACE MASK AND BCI BOX

FRONT VIEW · FACE SIDE OF THE MASK, AS DRAWN BY PIEEG

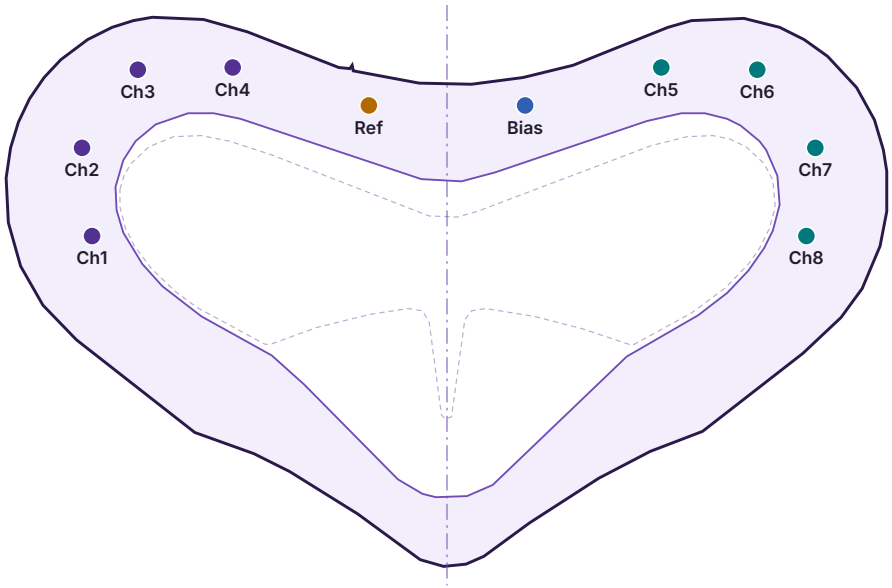
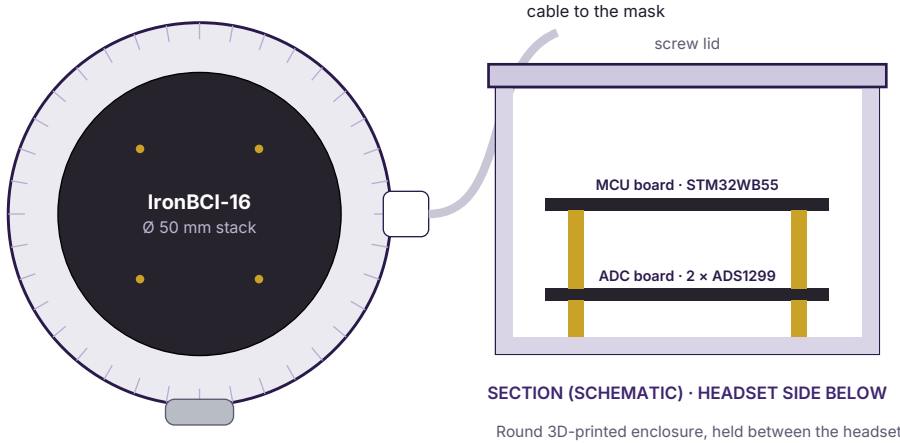


Figure 1 — Mask front view (face side) with the 10 electrode holes, traced from PiEEG's electrode map. Not to scale; the outline follows the headset model.

TOP VIEW · LID REMOVED (SCHEMATIC)



SECTION (SCHEMATIC) · HEADSET SIDE BELOW

Round 3D-printed enclosure, held between the headset and the rear head strap. M2 spacers between the boards.

Figure 2 — BCI box with the IronBCI-16 stack. Schematic, not to scale.

Title	PiEEG XR system layout
Doc. No.	XR-MD-002
Revision	Rev. A · 10 Nov 2026
Parts	Face mask · BCI box · IronBCI-16 module
Units	mm

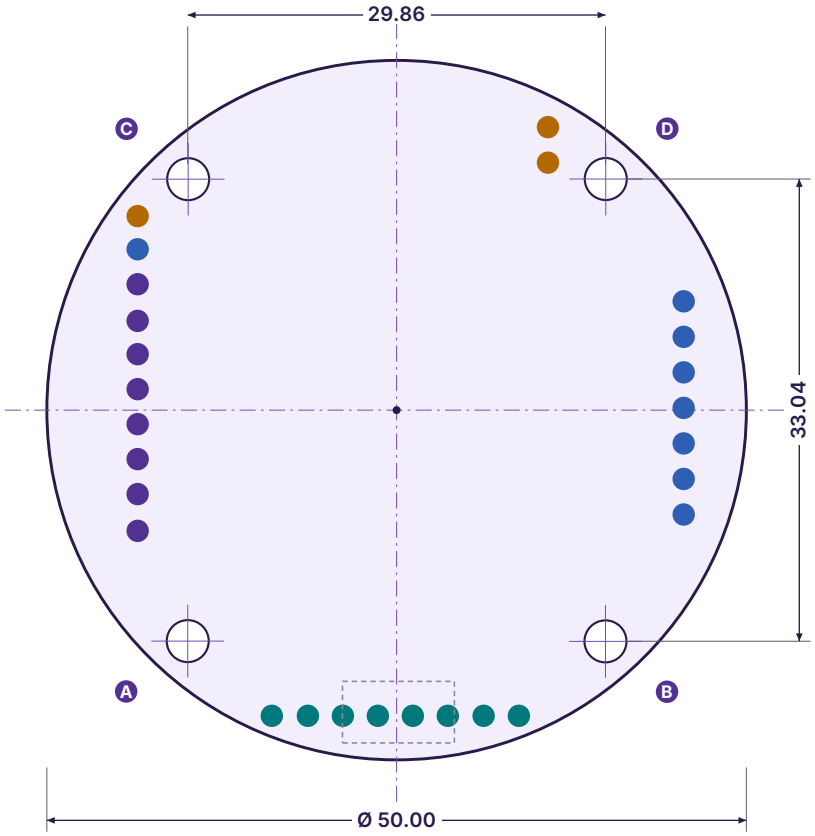
Notes

- Mask: silicone, 10 electrode holes Ø 1 mm; each mask is made for one headset model and clips in with 4 buckles (2 back, 2 front).
- BCI box: round 3D-printed enclosure, held between the headset and the rear head strap; USB-C 5 V input; cable to the mask.
- Mask and enclosure dimensions depend on the headset model and are not part of this drawing.

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02 IRONBCI-16 MODULE OUTLINE

TOP VIEW · ADC-BOARD ORIENTATION (STACK REFERENCE)



- J3.3 electrode header E1–E8 (1 × 10)
- BIAS pin
- REF pin / 2-pin power header
- J3.4 electrode header E9–E16 (1 × 8)
- 7-pin board-to-board header
- 4 × Ø3.0 holes, 29.86 × 33.04
- Micro-USB on the MCU board (above)



2 MCU board — STM32WB55, Micro-USB, BLE

1 ADC board — 2 × ADS1299, electrode headers

Exploded side view (schematic). M2 × 3 mm spacers with a washer at each board face.
Board-to-board: 7-pin signal header and 2-pin power header.

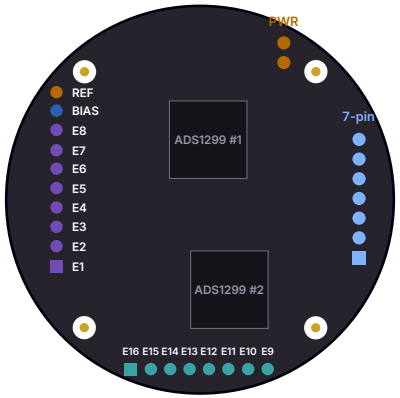
Module	IronBCI-16 · ADC + MCU boards
Datum	Board centre
Scale	1.85 : 1 (A4 print)

Notes

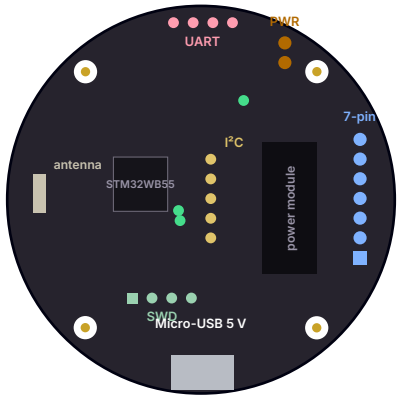
- Both boards Ø 50.0 mm, 4 holes at (±14.93, ±16.52) from the centre. An enclosure needs an inner diameter above 50 mm plus clearance.
- Board thickness 1.6 mm nominal per board (confirm before designing a new case).

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03 BOARD VIEWS AND INTERFACE LOCATIONS



ADC board (bottom)



MCU board (top)

Figure 4 — Both boards seen from above, in stack orientation. Pin 1 shown square. In PiEEG XR the mask cable lands on J3.3 and the box's USB-C socket feeds the 5 V input.

Board	Interface	Position (stack view, mm)	Use in PiEEG XR
ADC	**J3.3 electrode header**	X −18.51, Y −8.63 (E1) … +13.87 (REF)	Mask cable: Ch1–Ch8, Bias, Ref
ADC	**J3.4 electrode header**	Y −21.85, X −8.92 (E16) … +8.74 (E9)	Channels 9–16 (XR-16 only)
Both	**Board-to-board headers**	X +20.52 (7-pin) · X +10.82 (2-pin)	Internal SPI link and power
MCU	**5 V input**	bottom edge	Wired to the USB-C socket of the box
MCU	**SWD header**	see XR-PIN-004	Firmware: RST, SWDIO, GND, SWCLK
MCU	**Chip antenna**	left edge	Keep metal and the strap buckle clear

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04 FITTING TO THE HEADSET

Face mask

1 — Remove
Take off the headset's original facial interface.

2 — Align
Line up the four buckles of the PiEEG XR mask with the headset.

3 — Back buckles
Press to snap in the two back buckles.

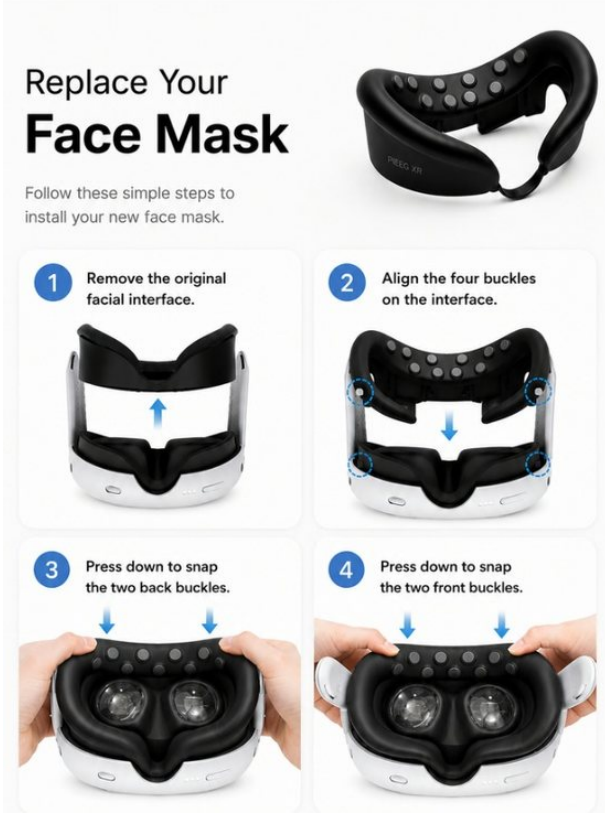
4 — Front buckles
Press to snap in the two front buckles. The mask must sit firmly, with no gaps or bending.

BCI box

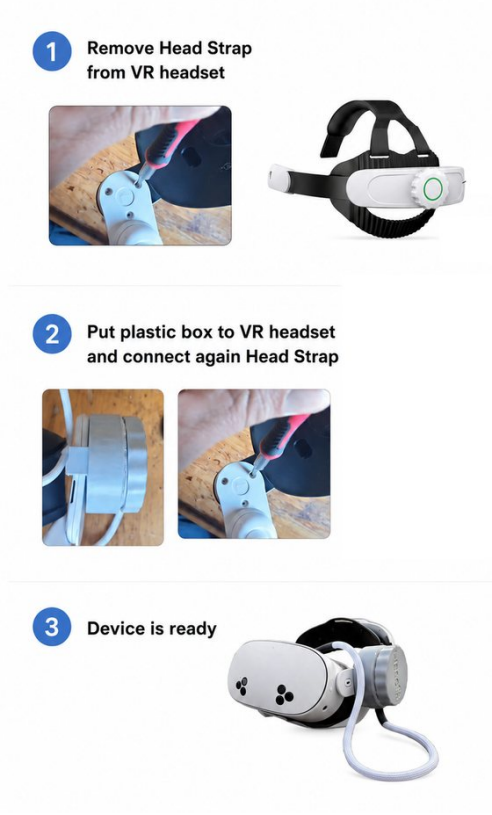
1 — Remove the strap
Detach the rear head strap from the headset.

2 — Place the box
Hold the box against the headset and reattach the strap so it clamps the box.

3 — Route the cable
Run the mask cable to the box; fix the power cable to the strap.



Replacing the face mask.



Mounting the BCI box.

NOTE Prefer your own mask?
Remove only the silicone insert from the PiEEG XR mask and fit it into the headset's original facial interface. Adjust until it sits firmly with no gaps or bending.

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